

# MUHAMMAD ABDULLAH GHANI

Software Engineer | Generative AI & MLOps Specialist

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## PROFESSIONAL SUMMARY

Results-driven Software Engineer specializing in Generative AI, LLM fine-tuning, RAG, Agentic AI, and production MLOps. Architected scalable microservices and autonomous AI systems that delivered 12% accuracy gains, 80% reduction in manual intervention, and zero-downtime deployments on Kubernetes and Docker.

Proven ability to turn complex data challenges into high-impact solutions for FinTech forecasting, SaaS customer support, and self-healing infrastructure.

## TECHNICAL SKILLS

**Languages:** Python, C++, JavaScript (Node.js), SQL

**AI/ML:** PyTorch, Hugging Face Transformers (BERT, DeBERTa, Llama), RAG, LLMs, LSTM, ResNet18, Scikit-learn, LangGraph

**MLOps/DevOps:** Docker, Kubernetes, GitHub Actions, FastAPI, CI/CD, Prometheus, Loki, Microservices

**Backend:** Flask, React.js, Tailwind CSS, Nginx

**Databases & Cloud:** PostgreSQL, MongoDB, Azure AKS

## EDUCATION

### B.S. Software Engineering

National University of Computer and Emerging Sciences, Pakistan

Aug 2022 - Present

## PROJECTS

### Autonomous Self-Healing Infrastructure (Agentic AI & DevOps)

*Python • LangGraph • Kubernetes • Docker • FastAPI • PostgreSQL • Azure*

- Designed LangGraph-based Agentic AI system that autonomously detected and resolved microservice failures in real-time, reducing manual intervention by **80%** across Kubernetes clusters.
- Integrated Prometheus and Loki for AI-driven anomaly detection and automated recovery, significantly minimizing downtime on Azure AKS.
- Implemented secure PostgreSQL audit trails ensuring **100%** data privacy compliance and full traceability of all AI decisions.

### FinTech Adaptive Forecasting System

*Python • MLOps • Docker • Kubernetes • LSTM/Transformers • CI/CD*

- Architected microservices platform that continuously retrains LSTM and Transformer models on live market data, delivering precise real-time trend predictions for investment decisions.
- Built end-to-end MLOps pipeline with automated drift detection (MAE/RMSE) and interactive candlestick visualizations, enhancing production model reliability.
- Deployed via Docker + Kubernetes with GitHub Actions CI/CD, achieving zero-downtime updates and high-availability scalability.

### ILM-ORA: AI-Powered Career & University Recommendation System (FYP)

*FastAPI • BERT • MERN • Scikit-learn*

- Led scalable FastAPI microservices integrating HEC and alumni datasets to deliver personalized university rankings and career guidance to thousands of users.
- Fine-tuned BERT for Aspect-Based Sentiment Analysis, increasing prediction accuracy by **12%** over baseline and improving recommendation relevance.
- Developed regression models predicting salary ranges and visualized career trajectories, providing students with actionable data-driven insights.

### QuickChat AI: Cloud-Based GenAI Customer Support Chatbot (SaaS)

*LLMs • RAG • FastAPI • Python*

- Engineered RAG system with domain-specific LLM fine-tuning to deliver accurate, business-contextual responses from uploaded knowledge bases for small-business clients.
- Built dynamic knowledge-retrieval pipelines and scalable cloud deployment, ensuring consistent tone and fast response times in live SaaS environments.
- Delivered production-ready multi-tenant chatbot supporting zero-manual-intervention operation